QUIZ 6 (Dated: 12-11-2024)
Data Structures and Algorithm Analysis

Roll No: ______________________ **Name: ____________________________**

Q1. Write a recursive function to determine whether it's possible to partition an integer array into exactly two parts such that:

1. Both parts collectively include all elements of the array (with no overlap and no elements left out).
2. The sum of elements in each part is equal.

If such a partition exists, return the index of the first element in the right-hand part. Otherwise, return 0.

```
input = 1, 2, 3, 4, 6
output = 0
input = 1, 1, 1, 1, 1, 2, 1
output = 4
input = 2, 3, 4, 6
output =: 0
input: 1, 2, 1, 2, 2, 4, 5, 7
output= 6
```

Q2. Given an array of positive integers, write a recursive function to determine whether it is possible to divide the array into multiple sub-arrays such that each sub-array has the same sum of elements. Each sub-array must have consecutive and non-overlapping elements. You should return true if such a division is possible and false otherwise.

```
i nput: 1, 4, 2, 3, 5, 5
Output: true
Input: 2, 2, 2, 2
Output: true
Input: 1, 2, 3, 4, 5
Output: false
```